Pricing Derivatives Using Lattice Tree Algorithm

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1 Introduction

After we interpret the Monte Carlo simulation for pricing the derivative (path-independent and path-dependent options), now I am going to introduce another famous method called lattice tree methods. Lattice tree methods are a well-known approach based on the binomial model. Based on a particular probability of the stock price going up or down, these methods use a backward approach to approximate the price of the derivatives. Therefore, in the following section, I would like to introduce two models: the binomial tree and the trinomial tree.

2 Binomial Tree

Based on the Black-Scholes model, we know that for the future value $V_T(S_T)$, the initial value is given by

$$V_0 = e^{-rT} E^Q[V_T(S_T)],$$

where Q stands for the risk-neutral probability measure. The stock price S_T satisfies the stochastic differential equation

$$\frac{dS_t}{S_t} = (r-q)\,dt + \sigma\,dW_T^Q,$$

where $W_T \sim N(0,T)$. In other words,

$$S_T = S_0 e^{\left(r-q-\frac{\sigma^2}{2}\right)T + \sigma W_T^Q}.$$

Hence, take the growth factors to be $u = e^v$ and $d = e^{-v}$, then denote $X_i =$ either v or $-v$ such that for n periods we have

$$S_T = S_0 e^{X_1 + X_2 + \cdots + X_n} = S_0 e^{\sum_{i=1}^n X_i}.$$

We know that the Brownian motion follows a normal distribution; therefore, we have the mean $\mu = (r-q-\sigma^2/2)T$ and variance $\sigma^2 T$.

Revisiting the formula derived earlier, we have

$$E[X] = E[X_1 + X_2 + \cdots + X_n] = nE[X_i],$$

where the X_i are independent and identically distributed.

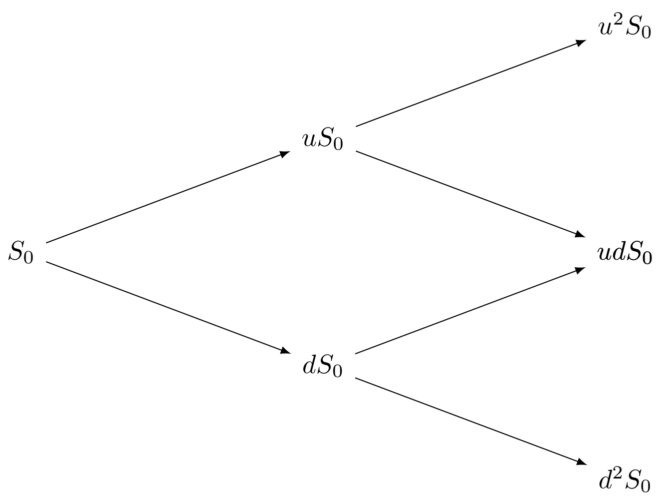

Figure 1: Binomial Tree for Two Periods

We have

$$(r - q - \sigma^2/2)T = nE[X_i] = nv(q_u - q_d),$$

where

$$q_u = \frac{e^{r\Delta t} - d}{u - d}, \quad u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}.$$

Hence, take the growth factors to be $u = e^v$ and $d = e^{-v}$. Then, denote X_i as either v or $-v$ such that for n periods, we have

$$S_T = S_0 e^{X_1 + X_2 + \cdots + X_n} = S_0 e^{\sum_{i=1}^{n} X_i}.$$

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(since the X_i are independent and identically distributed).

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$$\left(r - q - \frac{\sigma^2}{2}\right)T = nE[X_i] = nv(q_u - q_d),$$

3 Trinomial Tree

Now, let us move on to the trinomial tree. In reality, stock prices do not always only go up or down; sometimes they remain unchanged. Therefore, in the approximation of derivatives, we allow the stock price to go up, remain unchanged, or go down. To calculate the probabilities of these movements, we introduce a set of constraints.

Define the growth factor as

$$v = \lambda\sigma\sqrt{\Delta t},$$

where q_u, q_m, and q_d represent the probabilities of the stock price going up, remaining unchanged, and going down, respectively, satisfying

$$q_u + q_m + q_d = 1.$$

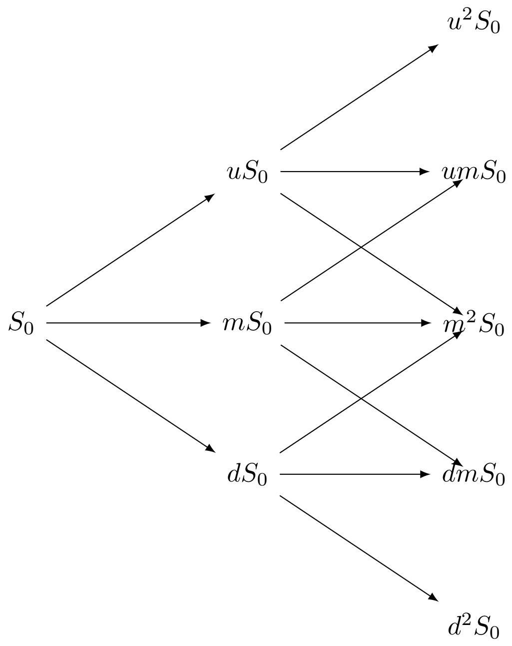


Figure 2: Trinomial Tree for Two Periods

Hence, we solve the system using algebraic methods to determine these probabilities.

In a trinomial tree model, we define the up, middle, and down factors for the stock price S at time step Δt as follows:

$$u = e^{\lambda\sigma\sqrt{\Delta t}}, \quad d = e^{-\lambda\sigma\sqrt{\Delta t}}, \quad m = 1, \tag{1}$$

$$S_u = Se^{\lambda\sigma\sqrt{\Delta t}}, \quad S_m = S, \quad S_d = Se^{-\lambda\sigma\sqrt{\Delta t}}, \tag{2}$$

where λ is a scaling parameter, σ is the volatility, and Δt is the time step.

The risk-neutral condition ensures the expected stock price matches the risk-free growth:

$$q_u u + q_m m + q_d d = e^{r\Delta t}, \tag{3}$$

where q_u, q_m, and q_d are the probabilities of the up, middle, and down moves, respectively, and r is the risk-free rate.

To match the variance of the stock price, we impose:

$$q_u(u - e^{r\Delta t})^2 + q_m(m - e^{r\Delta t})^2 + q_d(d - e^{r\Delta t})^2 = \sigma^2\Delta t. \tag{4}$$

Substituting u, m, and d, and letting $v = \lambda\sigma\sqrt{\Delta t}$, we rewrite the equations:

$$q_u e^v + q_m \cdot 1 + q_d e^{-v} = e^{r\Delta t}, \tag{5}$$

$$q_u(e^v - e^{r\Delta t})^2 + q_m(1 - e^{r\Delta t})^2 + q_d(e^{-v} - e^{r\Delta t})^2 = \sigma^2\Delta t. \tag{6}$$

Additionally, the probabilities must sum to one:

$$q_u + q_m + q_d = 1. \tag{7}$$

Using the substitution $v = \lambda\sigma\sqrt{\Delta t}$, we derive:

$$v(q_u - q_d) = \left(r - \frac{\sigma^2}{2}\right)\Delta t, \tag{8}$$

$$v^2(q_u + q_d) = \sigma^2\Delta t. \tag{9}$$

These equations, along with $q_u + q_m + q_d = 1$, form the system:

$$\begin{pmatrix} v & 0 & -v \\ v^2 & 0 & v^2 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} q_u \\ q_m \\ q_d \end{pmatrix} = \begin{pmatrix} \left(r - \frac{\sigma^2}{2}\right) \Delta t \\ \sigma^2 \Delta t \\ 1 \end{pmatrix}. \tag{10}$$

The determinant of the coefficient matrix is:

$$\det \begin{pmatrix} v & 0 & -v \\ v^2 & 0 & v^2 \\ 1 & 1 & 1 \end{pmatrix} = -2v^3. \tag{11}$$

The inverse of the coefficient matrix is:

$$\frac{1}{-2v^3} \begin{pmatrix} -v^2 & v & -v^2 \\ 0 & -2v^3 & 0 \\ -v^2 & -v & v^2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2v} & \frac{1}{2v^2} & 0 \\ 0 & -\frac{1}{v^2} & 1 \\ -\frac{1}{2v} & \frac{1}{2v^2} & 0 \end{pmatrix}. \tag{12}$$

Solving for the probabilities:

$$\begin{pmatrix} q_u \\ q_m \\ q_d \end{pmatrix} = \begin{pmatrix} \frac{1}{2v} & \frac{1}{2v^2} & 0 \\ 0 & -\frac{1}{v^2} & 1 \\ -\frac{1}{2v} & \frac{1}{2v^2} & 0 \end{pmatrix} \begin{pmatrix} \left(r - \frac{\sigma^2}{2}\right) \Delta t \\ \sigma^2 \Delta t \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{\left(r - \frac{\sigma^2}{2}\right) \Delta t}{2\lambda\sigma\sqrt{\Delta t}} + \frac{\sigma^2 \Delta t}{2(\lambda\sigma\sqrt{\Delta t})^2} \\ 1 - \frac{\sigma^2 \Delta t}{(\lambda\sigma\sqrt{\Delta t})^2} \\ -\frac{\left(r - \frac{\sigma^2}{2}\right) \Delta t}{2\lambda\sigma\sqrt{\Delta t}} + \frac{\sigma^2 \Delta t}{2(\lambda\sigma\sqrt{\Delta t})^2} \end{pmatrix}. \tag{13}$$

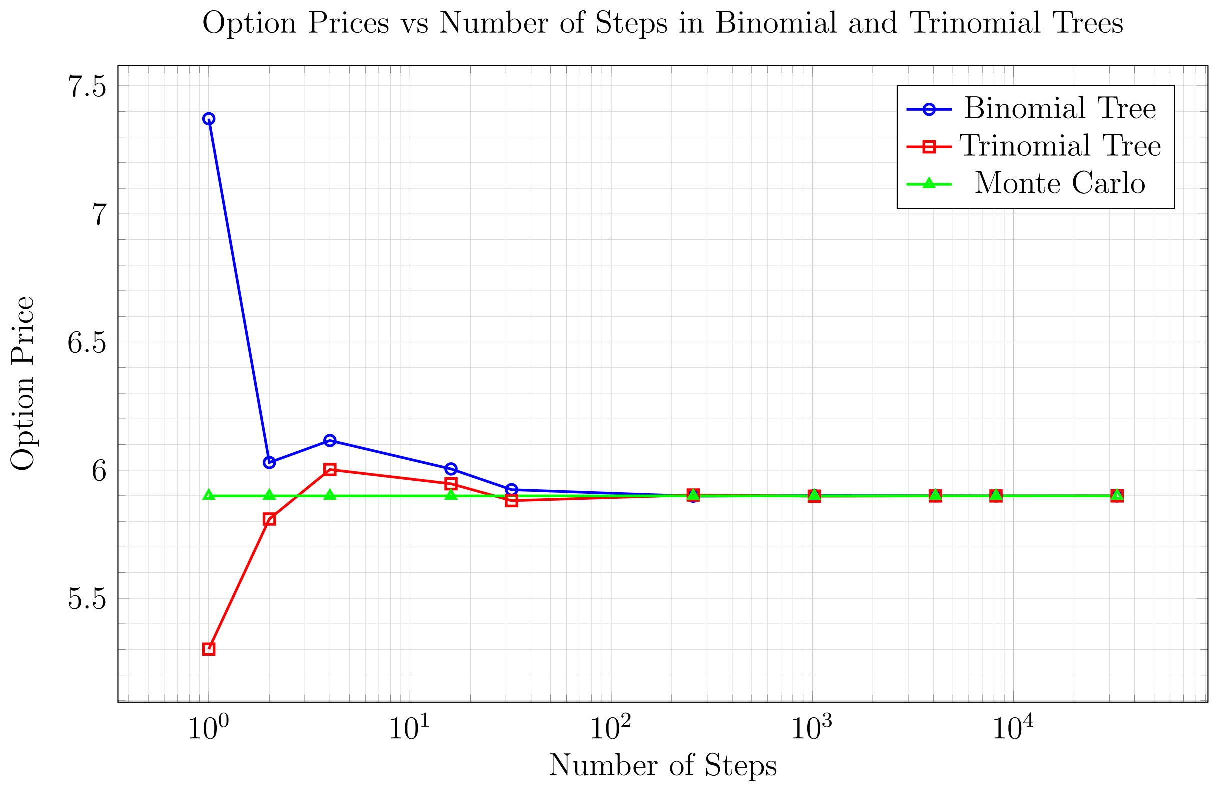
After simplification, the probabilities are:

$$\begin{pmatrix} q_u \\ q_m \\ q_d \end{pmatrix} = \begin{pmatrix} \frac{1}{2\lambda^2} + \frac{\left(r - \frac{\sigma^2}{2}\right)\sqrt{\Delta t}}{2\lambda\sigma} \\ 1 - \frac{1}{\lambda^2} \\ \frac{1}{2\lambda^2} - \frac{\left(r - \frac{\sigma^2}{2}\right)\sqrt{\Delta t}}{2\lambda\sigma} \end{pmatrix}. \tag{14}$$

4 accuracy of binomial tree and trinomial tree

Table 1: Option Prices vs Number of Steps in Binomial Tree and Trinomial Tree for call options

Number of Steps	Option Price (binomial)	Option Price (trinomial)
1	7.371480997587364	5.301105842746113
2	6.029562510328366	5.809088359588325
4	6.115485180344265	6.002052948604133
16	6.004493176763248	5.946550711991796
32	5.92349833627811	5.880679648505653
256	5.8982971363007435	5.902668177920129
1024	5.900152867676713	5.898193195493666
4096	5.899671179990084	5.899144273883466
8192	5.899153475524063	5.89911706474459
32768	5.899263319711965	5.899244559576235



At this stage, we assume $\lambda = \sqrt{3}$ for simplicity (I have no idea why $\lambda = \sqrt{3}$ and 3 is the kurtosis of the normal distribution). Also, we do not have the explicit formula for the option price; hence, we use the approximation of the price using the previous method (Monte Carlo simulation) and try to figure out the percentage error of the derivative.

Recently, we are using a European call option with the same scenario as a benchmark. Therefore, for the call option, the price is given by

$$\text{Price of call option} = 5.8992701404.$$

For $n = 32768$, we have

$$\text{price} = 5.899244559576235,$$

$$\text{percentage error} = |5.899244559576235 - 5.8992701404| = 0.00002558082,$$

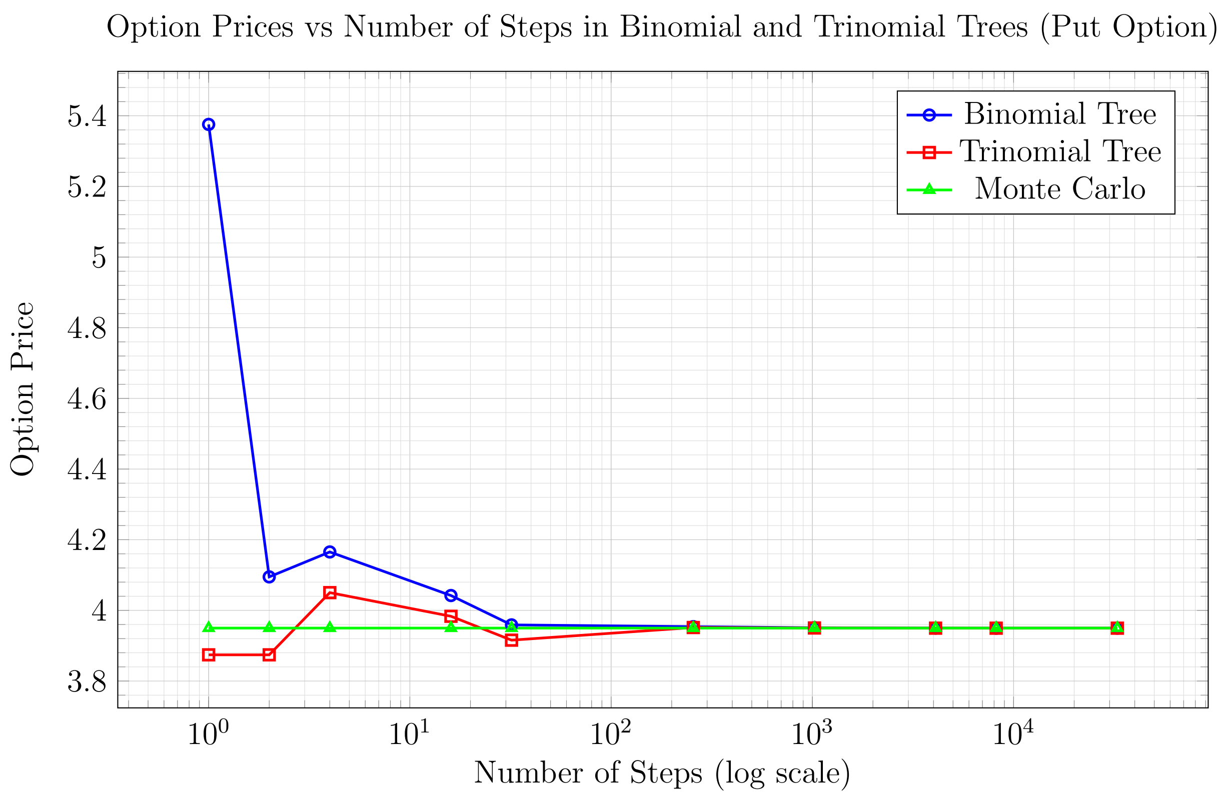
and the relative percentage error is

$$\frac{0.00002558082}{5.899244559576235} = 0.000433628\%,$$

which is an ideal error below our threshold of 0.1%. On the other hand , when the number of steps =32, the error is around 1%.

Table 2: Option Prices vs Number of Steps in Binomial Tree and Trinomial Tree (for put option)

Number of Steps	Option Price (binomial)	Option Price (trinomial)
1	5.375517178107498	3.8741357935313787
2	4.094782956324132	3.8741357935313787
4	4.165337859038444	4.050220988057417
16	4.041982255193202	3.9832677815419237
32	3.9588681299812287	3.9156428082909667
256	3.9538769861205982	3.9514556246832075
1024	3.9508251851542147	3.9501427449717137
4096	3.9498011334601353	3.949658581551729
8192	3.949549365293048	3.949590396266759
32768	3.949599732770323	3.949590396266759



Recently, we are using a European put option with the same scenario as a benchmark. Therefore, for the put option, the price is given by

$$\text{Price of put option} = 3.9498083425.$$

For $n = 32768$, we have

$$\text{price} = 3.949590396266759,$$

$$\text{percentage error} = |3.949590396266759 - 3.9498083425| = 0.00021794623,$$

and the relative percentage error is

$$\frac{0.00021794623}{3.949590396266759} = 0.0055182\%,$$

which is an ideal error below our threshold of 0.1%. On the other hand , when the number of steps =32, the error is around 1%.